to ensure that the next time this pathogen enters the body, it is eliminated far quickly. Once, the body is able to fight against the disease, it is believed to have built the immunity for it, also known as the body being immunized against the disease.

7.3.10 Hypersensitivity

Some of the individuals are very sensitive to some particles present in the environment. The exaggerated response of the immune system to certain antigens present in the environment is called **allergy (allo-altered, erg-reaction)**. The substances to which such an immune response is produced are called **allergens**. An allergen is an antigen that causes an allergic reaction. Allergic reactions begin within few seconds after the contact with the allergen and last about half an hour. The common examples of allergens are mites in dust, pollens and some proteins in insect venom. Hay fever and asthma are some common examples of allergy. **Symptoms** of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. Allergy is a form of over active immune response mediated by **IgE** and **mast cells.** It can also be due to the release of chemicals like histamine and serotonin from the mast cells.

Anaphylaxis is the classical immediate hypersensitivity reaction. It is a sudden, systematic, severe and immediate hypersensitivity reaction occurring as a result of rapid generalized mast-cell degranulation.

7.4 Immunodeficiency diseases

Immunodeficiency results from the failure of one or more components of the immune system. Primary immune deficiencies are caused by genetic developmental defects. Secondary immune deficiencies arise due to various reasons like radiation, use of cytolytic and immunosuppressive drugs and infections. **AIDS** is an acronym for Acquired Immuno Deficiency Syndrome. It is the deficiency of immune system, acquired during the life time of an individual indicating that it is not a congenital disease. AIDS is caused by Human Immuno Deficiency Virus **(HIV)**. It selectively infects helper T cells. The infected helper T cells will not stimulate antibody production by B-cells resulting in loss of natural defence against viral infection. On the basis of genetic characteristics and differences in the viral antigens, HIV is classified into the types 1 and 2 **(HIV-1, HIV-2)**.

Structure of HIV

The human immunodeficiency virus belongs to the genus **Lentivirus**. When observed under the electron microscope, HIV is seen as a spherical virus, **100-120 nm** in diameter, containing a dense core surrounded by a lipoprotein envelope. The envelope has **glycoprotein (gp)** spikes termed gp 41 and gp 120. At the core, there are two large **single stranded RNA**. Attached to the RNA are molecules of reverse transcriptase. It also contains enzymes like protease and ribonuclease. The core is covered by a capsid made of proteins. This is followed by another layer of **matrix proteins** as shown in the **Fig 7.18.**

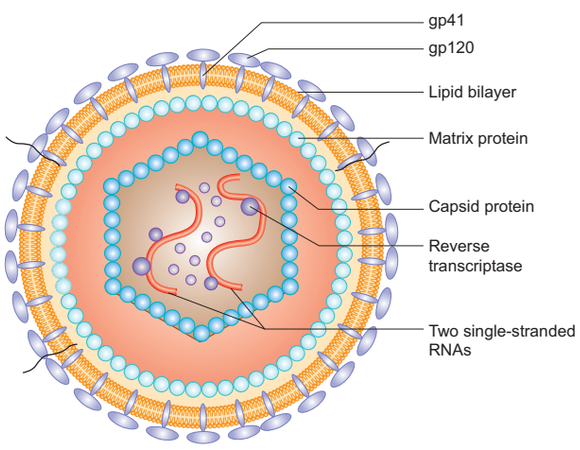


Fig. 7. 18 Structure of HIV

HIV Transmission

The HIV is often located within the cells especially in macrophages. HIV can

XII Std Biology-Zoology Chapter-7 EM.indd 127 23/12/2022 12:39:59